

Predicting Customer Satisfaction in E-commerce

Jose Castro

STAT310

Data Science for the Social Sciences

1 Introduction

Customer satisfaction is very important for online stores. Happy customers are more likely to buy again and recommend the store to others. Understanding what makes customers happy or unhappy can help improve delivery, products, and services. The goal of this project is to predict whether a customer will give a positive review (4-5 stars) or a negative review (1-3 stars) on Olist, a Brazilian e-commerce platform.

Olist connects small businesses from across Brazil to online marketplaces. When a customer places an order, the seller is notified to fulfill it. Once the product is delivered, the customer receives a satisfaction survey by email, where they can provide a rating and optional comments about their purchase experience.

This project uses three models (Logistic Regression, Random Forest, and XGBoost) to make predictions and compares their performance using simple measures like accuracy, precision, recall, and AUC (area under the ROC curve). All models are evaluated on a separate test set to avoid biased performance.

2 Data

This project uses the Brazilian E-Commerce Public Data by Olist, which contains information on over 100,000 orders placed between 2016 and 2018 across multiple marketplaces in Brazil. The data allows us to see each order from different perspectives. Order details include status and delivery times. Payment information includes total amount, type, and number of payments per order. Product information includes category, price, and other attributes. Finally, customer reviews provide both a numeric score and optional comments.

2.1 Exploratory Data Analysis

An initial exploration of the dataset revealed that most customers gave positive reviews (Figure 1), indicating overall high satisfaction.

Delivery timing also appeared important. Delayed orders were generally associated with lower review scores, while on-time deliveries were associated with higher review scores (Figure 2).

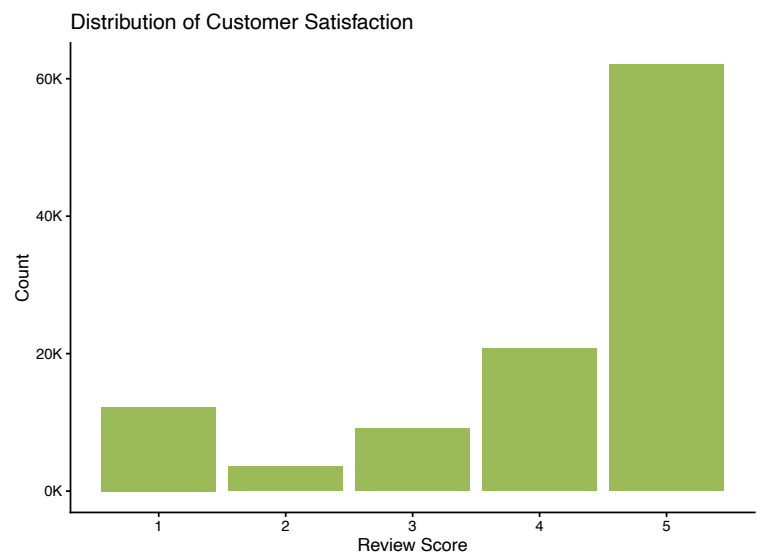


Figure 1



Figure 2

These observations suggested that delivery performance and other order-related factors could play an important role in predicting customer satisfaction.

3 Methods

Before building the models, the data was cleaned and prepared. This involved removing unnecessary identifiers and text fields, calculating total payment amounts and the number of payments per order, keeping only the latest review for each order, grouping rare product categories into an “others” category, creating new variables such as delivery delays and approval times, and defining the target variable, *positive_review*, as 1 for reviews of 4-5 stars and 0 for reviews of 1-3 stars. After merging and cleaning, the dataset included over 100,000 orders with variables describing the order process, product characteristics, payment information, and delivery performance.

An additional holdout set, representing 20% of the full dataset, was set aside at the start to evaluate final performance and ensure unbiased results. The remaining 80% of the data was then split so that 80% was used for training the models and 20% for initial testing during model development.

Three models were used to predict whether a review would be positive. Logistic Regression was used as a simple baseline model that estimates the probability of a positive review. Random Forest is an ensemble model that combines many decision trees to capture more complex patterns in the data. XGBoost is a gradient boosting method that often improves predictions, especially on complex datasets.

3.1 Challenges and Alternatives

Initial attempts to include all product categories without grouping resulted in sparse features, leading to reduced model performance. Grouping minor categories into “others” improved stability and performance.

4 Results

The performance of the models on the holdout set is summarized in the table below:

Model	Accuracy	Precision	Recall	AUC
Logistic Regression	0.7950846	0.8103614	0.9593135	0.6761771
Random Forest	0.8310689	0.8420638	0.9617739	0.7553761
XGBoost	0.8135405	0.8258001	0.9615338	0.7195506

Random Forest had the highest accuracy (83%) and precision (84%), making it the most reliable model at identifying positive reviews while minimizing mistakes. XGBoost performed slightly lower with an accuracy of 81% and precision of 83%. Logistic Regression had the lowest accuracy at 80% and an AUC of 0.68, showing that it was less effective at distinguishing between positive and negative reviews compared to the tree-based models. All models had a high recall (96%), meaning that most positive reviews were correctly identified.

4.1 Variable Importance

The most important factors influencing customer satisfaction varied across the models, reflecting the different ways each model captures relationship in the data.

Logistic Regression shows that delivery delays is the primary variable of whether a review is positive or negative (Figure 3). The model also suggests that customer state plays a meaningful role, though smaller than delivery delays, indicating that regional differences also affect review outcomes.

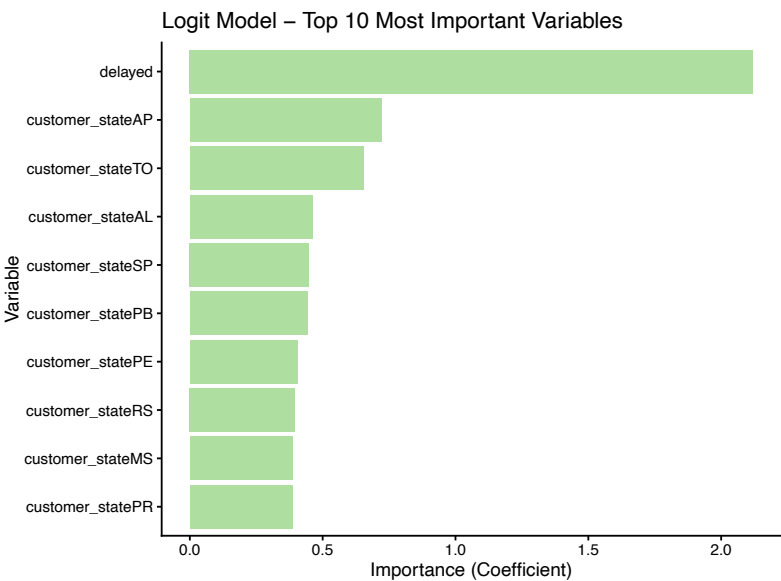


Figure 3

Random Forest highlights a broader set of influential variables. Delivery delays remains the most important variable, but there are other interesting variables like, total payment, product price, product weight, and product category. This suggests that in addition to delivery performance and location, financial and product characteristics also affect customer reviews (Figure 4).

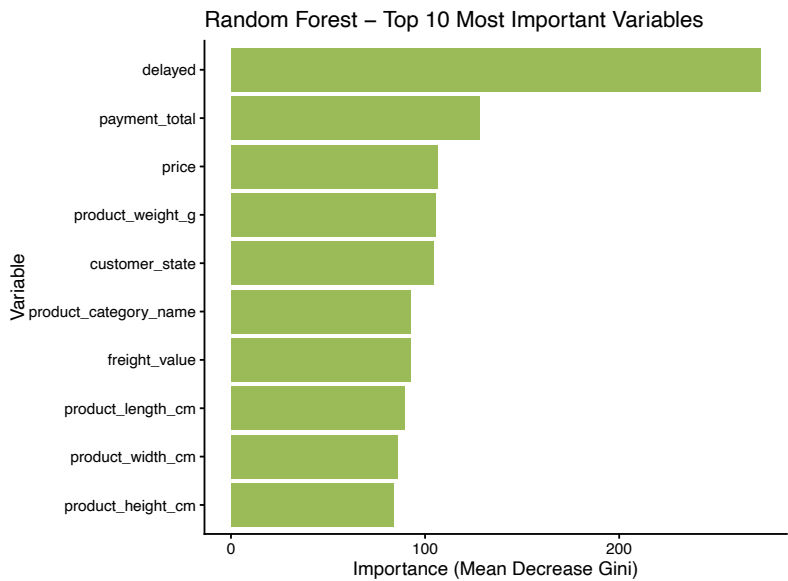


Figure 4

XGBoost also shows delivery delays at the top of its importance ranking, but it also emphasizes variables like total payment, product price, and approval hours, reflecting how this model captures more complex interactions between order timing and logistics when predicting customer satisfaction (Figure 5).

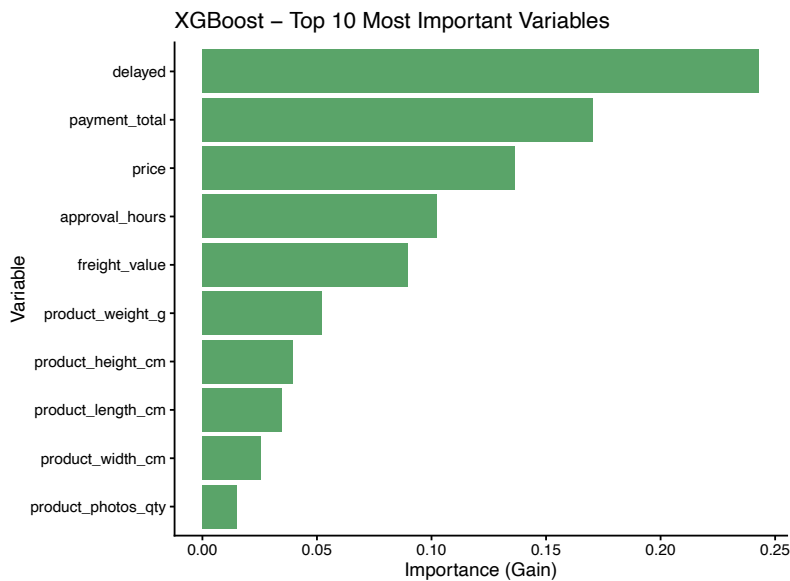


Figure 5

All models consistently identified delivery performance as a key determinant of satisfaction. However, tree-based models like Random Forest and XGBoost provided more detail by showing that several order and product features also matter, giving a clearer picture of what leads to positive customer experiences.

5 Conclusion

This project shows that customer satisfaction can be predicted using information about orders, products, payments, and delivery performance. Tree-based models such as Random Forest and XGBoost performed best because they were able to capture more detailed relationships between variables.

The results also shows that delivery delays, payment amounts, product price, and product characteristics are some of the most important factors affecting customer reviews. This information could help an e-commerce company identify which parts of the customer experience need the most attention. For example, reducing delivery delays, improving logistics, and focusing on products that tend to receive worse reviews could help improve customer satisfaction.

There are still some limitations. The models had more difficulty with rare product categories, which made some predictions less reliable. Adding more information, such as customer purchase history, repeat purchases, or loyalty measures, could improve the predictions and help companies better understand which customers are more likely to return or leave negative reviews.

This analysis shows that companies like Olist can use existing order and payment data not only to predict customer satisfaction, but also to make better business decisions about customer experience.

6 Works Cited

Brazilian E-Commerce Public Dataset by Olist. (n.d.). *Brazilian ecommerce public dataset*. Kaggle. <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

Does the optimal number of trees in a Random Forest depend on the number of predictors? (2013, March 5). *Cross Validated (Stack Exchange)*. <https://stats.stackexchange.com/questions/36165/does-the-optimal-number-of-trees-in-a-random-forest-depend-on-the-number-of-pred>

Olist. (n.d.). *Olist – marketplace for small businesses*. <https://olist.com>

XGBoost. (n.d.). *Parameters — XGBoost documentation*. <https://xgboost.readthedocs.io/en/stable/parameter.html>